Certified Recursive Bisection: Proof-Carrying Cuts for Algorithmic Redistricting

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Abstract

Recursive graph bisection offers a transparent construction rule for redistricting, but practical partitioners ordinarily return heuristic cuts without proving that a better permitted cut does not exist. We present *certified recursive bisection*, a proof-carrying formulation that preserves the enacted binary district-count tree while selecting one unique connected cut at every node. The objective is lexicographic: population deviation, weighted boundary cut, and canonical assignment. Split certificates bind parent and child unit universes into a complete tree whose one-seat leaves define the final plan. A bounded Rust implementation produces optimality and infeasibility artifacts, hostile tamper fixtures, RPLAN/RCTX packages, and deterministic pseudo-Boolean proof requests. A connected Rhode Island input with 25,649 Census blocks demonstrates data readiness but also identifies the unresolved scalability frontier: compact connectivity encoding, production discovery, and external proof generation and checking. The contribution is therefore a verified methods architecture, not a claim of nationwide exact readiness or superior political outcomes.

1 Introduction

Congressional redistricting combines a fixed numerical obligation—create exactly k districts—with a vast collection of geographically feasible assignments. Recursive bisection reduces that choice space to a sequence of two-way decisions. For California's 52-district delegation, the canonical schedule begins

$$52 \to 26/26, \qquad 26 \to 13/13, \qquad 13 \to 6/7.$$

The schedule is intelligible, reproducible, and naturally parallel. The remaining discretion lies in selecting each cut.

Practical graph partitioners such as METIS are well suited to finding strong balanced cuts quickly [4]. Their output, however, is not an optimality proof. Re-running a deterministic heuristic can establish what the program produced; it cannot establish that no better cut was permitted by the same rules.

This paper asks whether the recursive structure itself can become proof-carrying. Our answer is conditional but constructive. A governing rule must freeze the unit universe, population, adjacency and island links, split schedule, objective order, and tie-break. Software can then certify each required cut and bind the cuts into one recursive tree. This follows the procedural lesson of Huntington–Hill: mathematics settles execution only after the governing criterion has been selected [3].

The contribution is a methods and evidence contract. We do not claim that the chosen geographic objective is inevitable, that the resulting plan is a legal safe harbor, or that certified plans already outperform all alternatives on compactness, representation, or runtime.

2 Fixed Recursive Model

For a parent region requiring $k \geq 2$ districts, define

$$k_L = \left\lfloor \frac{k}{2} \right\rfloor, \qquad k_R = k - k_L.$$

Every unit is assigned to exactly one child, both children are nonempty and connected, and each child contains at least as many units as eventual districts. The solver may choose the cut; it may not alter (k_L, k_R).

Let P be parent population and P_L left-child population. The seat-ratio-scaled deviation is

$$\Delta_L = |kP_L - k_L P|\,.$$

The right-child expression is algebraically equal. We retain maximum and total deviation fields for schema continuity, although total deviation is twice the maximum for a binary split.

Among feasible assignments, the procedure minimizes lexicographically:

(1) maximum scaled population deviation;

(2) total scaled population deviation;

(3) weighted boundary cut; and

(4) the canonical binary assignment.

Equal-seat splits have label symmetry, removed by placing canonical unit zero in the left child. Unequal-seat splits are already oriented by the floor/ceiling seat counts; forcing unit zero left could exclude the ratio-correct solution and is therefore prohibited.

Island components use the project's deterministic rule: each non-main- component unit connects to its nearest same-county unit in the main component, with statewide fallback. Synthetic bridge edges receive the median land-boundary weight.

3 Split And Tree Certificates

A split instance binds its tree path, parent-certificate identity, canonical unit universe, populations, weighted edges, seat counts, and orientation rule. The result is either an optimal connected assignment or exact infeasibility. The bounded reference oracle commits to every candidate in deterministic order.

A whole-plan certificate must establish more than the validity of individual cuts. The verifier reconstructs both child contexts from each certified parent assignment. Submitted child populations, edges, and unit IDs are not trusted. Non-leaf nodes must appear in the breadth-first order of the canonical bisection schedule. One-seat leaves are ordered by binary path and must partition the root universe exactly once.

This construction distinguishes algorithmic proof from plan audit. An RPLAN audit can confirm assignment shape, population, contiguity, and provenance. The recursive certificate establishes the stronger model-scoped claim that no lexicographically better cut exists at each certified node.

The guarantee remains sequential. If a locally optimal parent cut produces a child that cannot be split under the same rules, the procedure fails explicitly; it does not silently backtrack to a worse parent cut.

4 From Discovery To Proof

Scalable certification separates two responsibilities. A fast discovery solver proposes a connected assignment and exact objective tuple. That record is an incumbent, not a proof.

Certification compiles three pseudo-Boolean decision problems:

1. Is any feasible cut better on population?
2. At the accepted population bound, is any cut better on boundary?
3. At both bounds, is any canonical assignment lexicographically earlier?

A satisfiable decision returns a counterexample and rejects the incumbent. Three independently checked unsatisfiability proofs establish the unique cut. The prototype emits OPB models intended for a proof-producing solver such as RoundingSat and an independent checker such as VeriPB [2, 1].

The initial compiler used static connectivity no-goods generated by bounded enumeration. The implemented `parent-depth-v1` alternative instead selects one root per child, one parent arc for each assigned non-root unit, and binary depths that increase strictly along selected arcs. This polynomial-size encoding has a committed RoundingSat boundary proof accepted by VeriPB.

The static compiler remains exponential in the number of units. When the incumbent already attains zero population deviation, the first lower-bound query is a trivial contradiction; the boundary and canonical stages remain the substantive optimality decisions.

5 Implemented Evidence

The bounded corpus establishes contract correctness, identity binding, and tamper resistance. It does not establish State-scale runtime. Rhode Island establishes that statutory-unit data and connectivity custody can be prepared; the current solver stops at the explicit 24-unit reference limit.

Artifact	Current evidence
Bounded split oracle	Optimal and infeasible certificates for equal and unequal seat ratios, with transcript tamper rejection.
Recursive tree	Parent-derived child contexts, canonical breadth-first nodes, and exact one-seat leaf coverage.
CLI package	Tree, RPLAN, RCTX, audit certificate, and manifest with tree-to-plan verification.
Hostile evidence	False optimum, ID tamper, node-order, missing-leaf, and leaf-universe substitutions are rejected.
Proof prototype	Static and compact OPB requests, a deliberate suboptimal SAT counterexample, and pinned RoundingSat/VeriPB population and compact-boundary proofs.
Rhode Island input	Connected 25,649-block RCTX with 66,097 land edges and 64 deterministic island bridges.

Table 1: Implemented evidence and its present claim boundary.

The State-scale frontier now goes one step further. Deterministic METIS plus nine articulation-safe boundary moves reaches Rhode Island populations 548,689 and 548,690. RoundingSat proves that deviation below 1 is impossible, and VeriPB accepts that proof. The boundary decision timed out, so the root cut is not yet fully certified.

The strongest demonstrated comparative result is therefore epistemic: certified BISECT can state what would count as proof, distinguish SAT counterexamples from UNSAT certification requests, and bind the final plan to the recursive evidence tree. Nationwide map-quality comparisons remain future empirical work.

On the committed path-8 root fixture, vendored METIS 5.1.0 with seed 42 selects the same assignment and objective as the certified oracle: $(\Delta_{\text{max}}, \Delta_{\text{total}}, \text{cut}) = (0, 0, 1)$ and assignment `00001111`. Certification does not improve that cut; it proves that the permitted answer is unique. A deliberate connected control `00011111` has population objective $(400, 800)$ and is correctly identified as worse. This single synthetic comparison cannot support national map-quality or runtime superiority.

6 What Certification Establishes

If all proof stages are independently accepted, certification establishes:

- the enacted recursive schedule was followed;
- every parent and child used the declared unit universe;
- no permitted connected cut improved the objective at any node; and
- the final assignment is the unique canonical BISECT plan.

Certification does not establish that the enacted axioms are normatively correct. It does not optimize partisan symmetry, competitiveness, communities of interest, racial representation, or every compactness measure. VRA and constitutional application remain separate legal questions.

Nor does the present implementation show that certified recursion "wins" on nationwide map quality. A defensible superiority study must run the same instances through certified and heuristic cut selection, report objective agreement and disagreement, measure runtime and proof size, and evaluate final plan metrics without selecting favorable states or seeds.

The current conclusion is narrower: relative to heuristic output, solver metadata, and RPLAN audit alone, certified recursive bisection provides the strongest model-optimality claim architecture currently implemented in the project. Whether that proof strength can be delivered at national scale is the central open engineering question.

7 Conclusion

Certified recursive bisection preserves the most recognizable feature of BISECT—a fixed sequence of proportional two-way cuts—while replacing unproved per-node choices with a unique, checkable result. Bounded implementation, hostile fixtures, package verification, and Rhode Island block custody show that the architecture is concrete rather than aspirational.

The remaining work is equally concrete: integrate the compact encoding and pinned proof toolchain at State scale, integrate an exact discovery solver, and obtain the first independently checked State proof. Until then, METIS remains the practical nationwide benchmark and certified BISECT remains the proof-oriented North Star.

References

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